

Mathematics for ML

- ① How machines represent relationships → functions
- ② How they handle scale & uncertainty → Logarithms
- ③ How they understand changes → Derivatives
- ④ How they improve over time → Rate of change

Age → 25
Salary → 30,000

Variable - abstraction layer
 $\{a, b, c\}$ placeholder for meaning
 $\{x = \text{experience}, y = \text{salary}\} \rightarrow \text{relationship begins}$
 $2x + 3, x^2, x + y, ax + b$

Function
 is a transformation
 It takes input → produces an output.
 deterministic relationship b/w input & output.
 $f(x) = y$ Prophet

- ① Linear relationship
 $f(x) = 2x + 5$
 $x \rightarrow$ input
 $y \rightarrow$ output
 unit change
- ② Non-linear relationship
 $f(x) = x^2$
 def $f(x)$: return x^{**2}
 print $f(4)$ → 16
 $f(\text{experience}) \rightarrow \text{salary}$
 $f(\text{user-behaviour}) \rightarrow \text{recommendation}$
 $f(x) = x^2$

Logarithms → Handle scale & uncertainty
 1000, 1000000
 control explosion.
 $10^3 = 1000 \rightarrow \log_{10} 1000 = 3$
 $\log_{10} 10^3 = \log_{10}(1000)$
 $3 \times 1 = \log_{10}(1000)$

Log transformation.

Original	Log
10	1
100	2
1000	3
1000000	6

 $\log_{10}$ } $\ln$
 Logarithm turned explosion into linear.
 $\log(ab) = \log(a) + \log(b)$
 Log in uncertainty (probability)

$\frac{90}{100} \times \frac{80}{100} \times \frac{70}{100} = \text{very small}$
 $0.9 \times 0.8 \times 0.7$
 after a lot of multiplication, almost zero.
 $P = P_1 \times P_2 \times P_3$ underflow error.
 $\log P = \log(P_1) + \log(P_2) + \log(P_3)$

Limit

approaching infinitely close.
 $\lim_{n \rightarrow a} f(n) = L$
 $f(a)$
 destination of behaviour, not the actual position.
 LHL RHL
 $x \rightarrow a^- \quad x \rightarrow a^+$
 $LHL = RHL$

$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$
 $\lim_{x \rightarrow 0} \frac{1}{x} = 0$
 $\lim_{n \rightarrow 0} \ln(1+n) = 0$
 impact number as $n \rightarrow 0$
 $n = \text{np.linspace}(-1, 1, 100)$
 $y = \text{np.sin}(x)$
 point $(y[10])$
 output → 10 mango / 10 people
 $x = \text{np.linspace}(0.0001, 1, 100)$
 $y = \text{np.sin}(x)$
 point $(y[15])$
 10 mango / 10 people
 0 mango / 0 people

Derivatives → Changes

What is your speed? 100 distance / time
 100 km / hr
 something is changing at a moment.
 $\frac{d}{dx} \rightarrow f(x) = x^2$
 $f'(x) = 2x$
 $\frac{d}{dx} x^n \rightarrow nx^{n-1}$
 $\frac{d}{dx} x^3 = 3x^2$
 $\frac{d}{dx} x^2 = 2x$
 $\frac{d}{dx} x = 1$
 at $x = 2$, slope → 4
 at $x = 1$, slope → 2
 Tangent line

$f(x) = x^2 \rightarrow f'(x) = 2x$
 Derivative → rate of change
 y changes when x changes
 Slope $x < 0$
 Slopes $x > 0$
 Driving → speed = derivative of position
 mk → gradient = derivative of loss

x	slope (2x)	meaning
-2	-4	steep downward
-1	-2	mild "
0	0	flat
1	2	rising
2	4	steep rising

 Derivative → measure the instantaneous rate of change in a function.
 $f(x) = x^2$
 derivative = $2x$
 at $x = 3 \rightarrow \text{slope} = 2 \times 3 = 6$
 Direction → increase or decrease parameter?
 Magnitudes → how aggressively should we change?
 Optimizations → where is the minimum error?
 loss f' → measure error.
 direction → tell slope
 gradient descent → update model